

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CS3103-1 –THEORY OF COMPUTATION
 (For CSE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **One** full question from each **Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is an application of Finite Automaton?
 A) Compiler Design B) Grammar Parsers
 C) Text Search D) All of the mentioned
- In NFA, this very state is like dead-end non final state:
 A) ACCEPT B) REJECT
 C) DISTINCT D) START
- Which among the following can be an example of application of finite state machine (FSM)?
 A) Communication Link B) Adder
 C) Stack D) None of the mentioned
- The number of tuples in an extended Non-Deterministic Finite Automaton:
 A) 5 B) 6
 C) 7 D) 4
- A finite automaton recognizes
 A) Any language B) Context sensitive language
 C) Context-free language D) Regular language
- NFA, in its name has 'non-deterministic' because of :
 A) The result is undetermined B) The choice of path is non-deterministic
 C) The state to be transited next is non-deterministic D) All of the mentioned
- In the context of a finite automaton, the extended transition function is denoted by:
 A) $\delta(q, a)$ B) $\delta^*(q, w)$
 C) q_0 D) Q'
- $\epsilon + RR^* = ?$
 A) R B) R^+
 C) R^* D) ϵ
- Which of the following statement is correct?
 A) All Regular grammar are context free but not vice versa B) All context free grammar are regular grammar but not vice versa
 C) Regular grammar and context free grammar are the same entity D) None of the mentioned
- A CFG is ambiguous if
 A) It has more than one rightmost derivations B) It has more than one leftmost derivations
 C) No parse tree can be generated for the CFG D) Both (A) and (B)

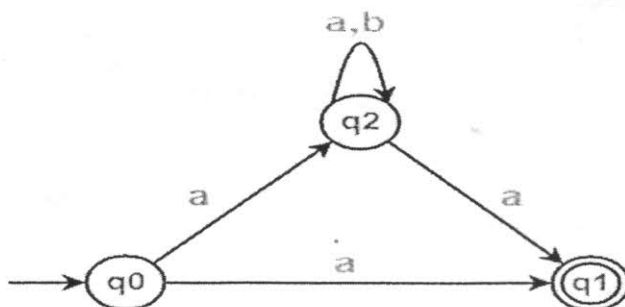
11. Which of the strings do not belong to the regular expression $(ba + baa)^*aaba$
 - A) baaaba
 - B) babaabaaaba
 - C) babababa
 - D) baaaaba
12. What is the primary purpose of the State Elimination Method?
 - A) To minimize the number of states in a finite automaton
 - B) To convert a finite automaton into a regular expression
 - C) To convert a regular expression into a finite automaton
 - D) To remove unreachable states from the finite automaton
13. Which of the following statements is FALSE about epsilon closure?
 - A) Epsilon closure of a state in an NFA may include more than one state.
 - B) Epsilon closure is computed only for DFA.
 - C) Epsilon closure can be used to transform an NFA with epsilon transitions into an equivalent DFA.
 - D) The epsilon closure of a state always contains the state itself.
14. Which of the following grammars is ambiguous?
 - A) $S \rightarrow aSa \mid bSb \mid \epsilon$
 - B) $S \rightarrow AB \mid a$
 - C) $S \rightarrow SS \mid a$
 - D) $S \rightarrow a \mid b$
15. Which of the following grammars is in Chomsky Normal Form?
 - A) $S \rightarrow aS \mid b$
 - B) $S \rightarrow AB \mid a$
 - C) $S \rightarrow aA \mid bB, A \rightarrow a, B \rightarrow b$
 - D) $S \rightarrow a \mid b$
16. If the PDA does not stop on an accepting state and the stack is not empty, the string is:
 - A) rejected
 - B) goes into loop forever
 - C) all of the mentioned
 - D) none
17. Halting state of Turing machine are:
 - A) Start and stop
 - B) Accept and reject
 - C) Start and reject
 - D) Reject and allow
18. The value of n if Turing machine is defined using n -tuples:
 - A) 6
 - B) 7
 - C) 8
 - D) 5
19. In one move the Turing machine:
 - A) May change its state
 - B) May write a symbol on the cell being scanned.
 - C) May move the head one position left or right
 - D) All of the above
20. The Turing machine is the machine format of _____ language.
 - A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA):



8 L3 1 1

- b) Define the following terms with an example:
 i) NFA ii. Regular Language iii. Length of a string
 iv. Power of an alphabet
2. a) Define DFA. Design DFA for the following languages:
 i. $L(M) = \{w \in \{a,b\}^* : w \text{ does not contain three consecutive b's}\}$
 ii. the language that accepts binary numbers that are divisible by three.
- b) Write differences between DFA, NFA and ϵ -NFA.
- c) Let $L = \{0^k 1^{2k} : k \in \mathbb{N}\}$. Prove that L is not regular.
3. a) Define regular expression. Write the regular expression for the following languages:
 i) starting with a but not having consecutive b's.
 ii) all the string do not contain the substring 01.
 iii) string should have atleast one 0 and atleast one 1.
 iv) string in which every 0 is immediately followed by 11.
- b) Consider the following ϵ -NFA
1. Compute ϵ -closure of each state
 2. Convert the automation to a DFA.

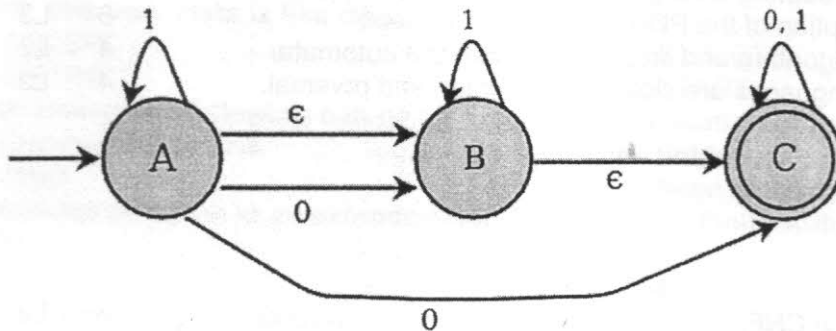
8 L1 1 1

8 L3 1 1

4 L1 1 1

4 L3 2 1

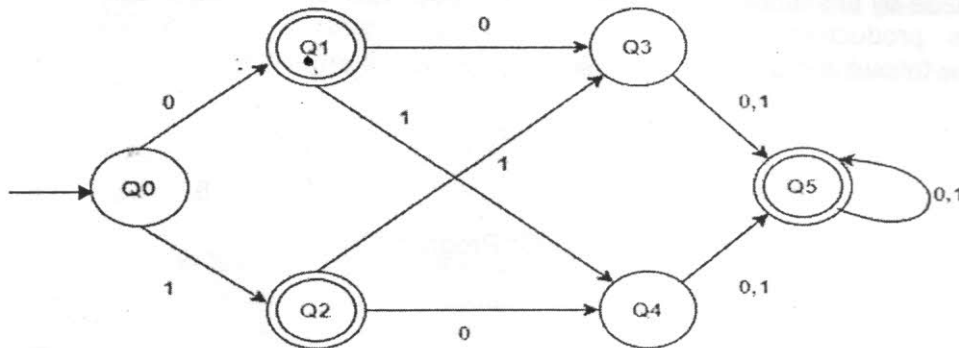
8 L3 2 1



8 L3 1 1

Unit – II

4. a) Define distinguishable and indistinguishable states. Minimize the following DFA using table filling algorithm.



8 L3 2 1

- b) What is ambiguous grammar? Check whether the given grammar G is ambiguous or not for $w = abba$

 $S \rightarrow SS$
 $S \rightarrow a$
 $S \rightarrow b$

4 L3 3 1

- c) Define PDA? Explain Acceptance by final state and Acceptance by Empty stack. 4 L2 4 1
5. a) Define Context-free grammar. Give context-free grammars that generate the following languages.
 (a) $L = \{ w \in \{0, 1\}^* \mid w \text{ contains at least three 1s} \}$
 (b) $L = \{ w \in \{0, 1\}^* \mid w = w^R \text{ and } |w| \text{ is even} \}$
 (c) $L = \{ w \in \{0, 1\}^* \mid \text{the length of } w \text{ is odd and the middle symbol is } 0 \}$
 (d) $L = \text{set of strings over the alphabet } \{a, b\} \text{ with } a\text{'s more than } b\text{'s.}$ 8 L3 3 1
 b) Design a PDA for accepting a language $\{a^n b^n \mid n \geq 1\}$. 4 L3 4 1
 c) Consider the grammar G with production:
 $S \rightarrow AbB$
 $A \rightarrow aA \mid \epsilon$
 $B \rightarrow aB \mid bB \mid \epsilon$
 Obtain leftmost derivation, rightmost derivation, and parse tree for the string "aabab". 4 L2 3 1
6. a) Design a PDA for accepting a language $\{a^n b^{2n} \mid n \geq 1\}$ and show the instantaneous description of the PDA on input "aaabbbbb". 8 L3 4 1
 b) Explain table filling algorithm and steps to minimize the automata. 4 L2 3 1
 c) Prove that regular languages are closed under union and reversal. 4 L2 3 1

Unit – III

7. a) Consider the grammar
 $S \rightarrow AAA|B$
 $A \rightarrow aA|B$
 $B \rightarrow \epsilon$
 Obtain the grammar in CNF. 8 L3 3 1
 b) With a neat diagram, explain the working of a basic Turing machine. Design a Turing machine to accept $L = \{ww^R \mid w \in (a+b)^*\}$. 8 L3 5 2
8. a) Design a Turing machine to accept the language $L = \{0^n 1^n 2^n \mid n \geq 1\}$. Show the moves made by this Turing machine for the string 001122. 8 L3 5 1
 b) What are useless productions? Eliminate ϵ , unit and useless productions from the following grammar:
 $A \rightarrow bA \mid Bba \mid aa$
 $B \rightarrow aBa \mid b \mid D$
 $C \rightarrow CA \mid AC \mid B$
 $D \rightarrow a \mid \epsilon$ 8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
